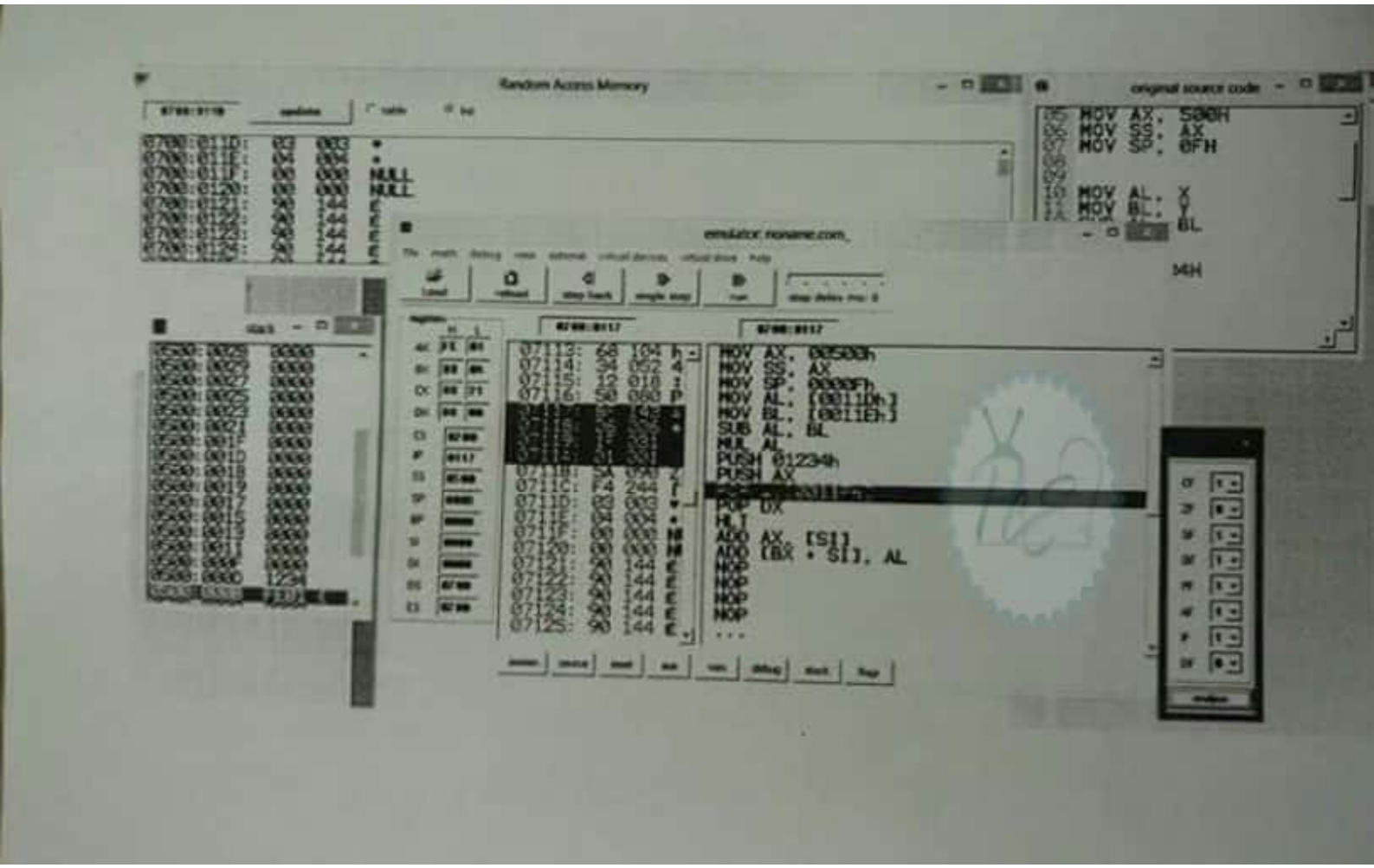


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Second Question: Complete the following program according to the notes below:

```
    ORG 1000H
    JMP ..................... ;(1)

    line db 10, ?, 10 dup('-')... ;(2) define a new line char
    buff db 10, ?, 10 dup(' ')
    StS db 'Smooky'
    DtS db ..............  dup (' ') ;(3)

begin:
;enter a single char. from the K.B.
    MOV AH, ..... ;(4)
    INT 21H

;print a new line on screen - by calling procedure printline
    ..............  printline ;(5)

;print number 9 on screen
    MOV AH, ..................  ;(6)
    MOV DL, ..................  ;(7)
    INT 21H
    CALL printline

;enter "C++" from the K.B. into buff
    MOV AH, ..................;(8)
    MOV DX, ..................;(9)
    INT 21H

;copy the contents of StS to DtS 1 byte at a time
;starting from the end of the strings

    STD
    MOV CX, ..................;(10)
    LEA SI, ..................;(11)
    LEA DI, ..................;(12)
    REP
    ..........................;(13)

;add $ at the end of DtS
    MOV DtS[.....], ..................;(14), (15)

    CALL printline

;print string DtS on screen
    MOV AH, 9
    LEA DX, ..................;(16)
    INT 21H

HLT

printline PROC
    MOV AH, ..............;(17)
    LEA DX, nL
    ..........................;(18)
    RET
printline ENDP
```

Comment Number	Value	Comment Number	Value
(1)		(10)	
(2)		(11)	
(3)		(12)	

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(4)		(13)	
(5)		(14)	
(6)		(15)	
(7)		(16)	
(8)		(17)	
(9)		(18)	

Third Question: According to the given figure on the next page, fill-in the blan correct answer:

1- The physical address of the next instruction to be exe

2- The length of the current instruction is: _____

3- Variable X's value is _____ and Y's value is _____

4- Variable X's offset is: _____

5- Variable Y's logical address is: _____

6- The current value of register AX is _____, and the value of after executing the program is _____

7- the value of SS is _____, and the current value of SP is _____

8- The starting value of SP was _____

9- The instruction(s) that affects the flags is/are:

_____

10- The value of variable Z after executing the whole program is: _____

11- The physical address of the variable Z is _____

12- Variable Z is defined using _____ directive, show how the stored in memory:

Logical Address	Value

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Computer Engineering department
Microprocessor Final Lab Exam

Name: _____ Date: _____

First Question: Fill in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10

Q1) The number of the data lines (A.L.) in the 8086 architecture is
1- 8 lines 2- 16 lines 3- 20 lines 4- 24 lines

Q2) The number of the address lines (D.L.) in the 8086 architecture is
1- 8 lines 2- 16 lines 3- 20 lines 4- 24 lines

Q3) The register that contains the starting address of the extra segment is
1- CS 2- DS 3- SS 4- ES

Q4) To define a variable called num that allocates two bytes in memory you use the _____ directive:
1- DW 2- DB 3- DD 4- All of them

Q5) To perform the multiplication process between 2 and 3, you use the instruction
1- MUL 2- IMUL 3- (1 and 2) 4- None of them

Q6) The size of the data segment equals to _____
1- 64MB 2- 64KB 3- 46MB 4- 46KB

Q7) The instruction JE performs the jump operation according to the value of _____ flag
1- EF 2- NEF 3- NZF 4- ZF

Q8) The instruction CLD, used to put the value _____ in _____ flag
1- (0) in OF 2- (1) in OF 3- (0) in DF 4- (1) in DF

Q9) To scroll-up the screen with a red background and a gray fore-color you use interrupt _____
1- 21h 2- 10h 3- 16h 4- All of them

Q10) The register that can be divided into two halves is
1- EX 2- DX 3- DS 4- (1) and (2)

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Faculty of engineering technology
Computer Engineering department

final assembly lab Exam(A)

2nd Semester 2017/2018

Student No: [redacted]
Student Name: [redacted]

instructor name: Manal al_rawashdeh
Mohammad al_otari

Exam time:50 min

➢ Answer the following Squestion

Question # 1: Write an assembly code to input number (from 0 to 9) from keyboard then print "even" if number is even ,other wise print "add" Write complete program?(10 pts)

Question #2: Trace the following (10 pts):

#1:

```
Org 100h
Mov ah,9
Lea dx,str
Int 21 h
Hlt
Str db " assembly",09,41h,42h,24h
```

[redacted]

#2:

```
org 100
Mov ah,9
Lea dx,x1
int 21H
hlt
X1 db "ab"
X2 db "cd5"
X3 db "ef5"
```

[redacted]

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#1

Mov ax ,3232h

Mov bx,2734h

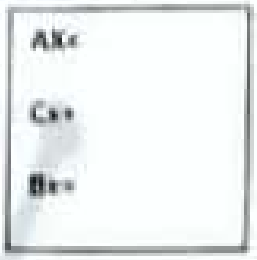
Push ax

Push bx

Pop cx

Pop bx

xchg



Question #3:

Choose the correct answer for following points: (14pts)

1. Assuming AL = 4 and BL = 5, the code:
   MOV AL, BL MOV BL, AL is equivalent to:
   a) MOV BL, AL
      MOV AL, BL
   c) MOV BL, 4
   d) MOV BL, 5
   e) XCHG AL, BL
2. Assuming AL = 1 and BL = -1, the instruction: MUL BL will produce:
   a) AX = 1 b) AX = -1 c) AX = 00FFh d) AX = FF00h e) AX = 0100h

3. Assuming AL = -1 and BL = -1, the instruction: DIV BL will produce:
   a) AX = 1 b) AX = -1 c) AX = 00FFh d) AX = FF00h e) AX = 0100h

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Faculty of engineering technology
Computer Engineering department

final assembly lab Exam(A)

Student No:
Student Name: [redacted]

2nd Semester 2017/2018
instruction name: Manal al [illegible]
Mohammed al [illegible]

Exam time:50 min

4. neg is equivalent to :
a)multi -1 b) Subtraction c) div d) add e)a+b

5. LEA is means
a) Load effective address b) Load address. c) Load physical address. d) None of them .

6. Push is means
a) Operand onto the top of the hardware supported register
b) places its operand onto the top of the hardware supported stack in memory
c) Operand onto the top of the hardware supported stack in memory
d) None above

7.test instruction is equivalent to:
a)cmp b)jnz c)shl d)shr e)all of them

Question #4:

Write an assembly code to exchange the values in the two arrays using stack (push and pop instructions)? (10)

Question #5:

Suppose that DS=0200H, BX=0300H, and DI=400H. Determine the memory address accessed by each of the following instructions, assuming real mode operation(6pts)

(a) MOV AL, [1234H]

(b) MOV AX, [BX]

(c) MOV [DI], AL

The End

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